

L^AT_EX Lab-Exam Complete Reference

Syntax, Symbols, Tables, Figures & Explanations — One Sheet

Personal exam documentation

July 21, 2026

Compile twice with pdflatex . Every symbol appears beside its code. Every construct is shown as code then output . Ctrl-F any feature during the exam.
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Contents

1 The 30-Second Preamble

Everything starts with a *document class* and a *preamble* (the zone before `\begin{document}` where you load packages). Copy one of these first, then start typing.

Minimal — math, images, colour, links:

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb,graphicx,xcolor,hyperref}
\begin{document}
% ... your content ...
\end{document}
```

Exam-standard — add only what the paper needs:

```
\documentclass[11pt]{article}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage[T1]{fontenc}\usepackage[utf8]{inputenc}
\usepackage{amsmath,amssymb,amsthm}      % maths + theorems
\usepackage{graphicx}                    % images
\usepackage[table,dvipsnames]{xcolor}    % colour + coloured tables
\usepackage{multirow,array,booktabs}    % strong tables
\usepackage{subcaption,wrapfig}         % subfigures, wrapped figures
\usepackage{enumitem}                   % custom list labels
\usepackage{listings}                   % code
\usepackage{authblk}                    % numbered author affiliations
\usepackage{hyperref}                   % LOAD LAST: links + clickable TOC
```

Golden rules

- ✓ `amsmath` for equations, `graphicx` for images, `xcolor` for colour, `hyperref` last.
- ✓ A *package* is loaded with `\usepackage{name}`; options go in brackets: `\usepackage[opt]{name}`.
- ✓ Comment out a line with `%`. Escape specials: `\% \\$ \& \# \_ \{ \}`.

2 Document Structure

2.1 Title block and authors

The `authblk` package wires each author to a numbered affiliation:

```
\title{My Paper Title}
\author[1]{Alice Smith}\author[2]{Bob Jones}
\affil[1]{University A, City}
\affil[2]{Institute B, City}
\date{} % empty {} deletes the date; omit it -> today's date
\maketitle % first line INSIDE \begin{document}
```

2.2 Sectioning

The commands, from largest to smallest, are `\section`, `\subsection`, `\subsubsection`, `\paragraph`, `\subparagraph`. Each auto-numbers and enters the Table of Contents. Add a star (`\section*{...}`) to make a heading *unnumbered* and keep it *out* of the TOC — useful for “Abstract” or an unnumbered “Circuit Model” heading.

2.3 Table of contents

`\tableofcontents` prints every heading with page numbers, and becomes clickable once `hyperref` is loaded. Because it reads data recorded on the *previous* run, you must compile **twice**.

3 Text Formatting, Fonts, Sizes and Colour

3.1 Weights and shapes

Output	Code	Output	Code
bold	<code>\textbf{..}</code>	<i>italic</i>	<code>\textit{..}</code>
<u>under</u>	<code>\underline{..}</code>	mono	<code>\texttt{..}</code>
SMALL CAPS	<code>\textsc{..}</code>	<i>emphasis</i>	<code>\emph{..}</code>
sans serif	<code>\textsf{..}</code>	roman	<code>\textrm{..}</code>

3.2 Size ladder

From tiny to huge (declared inside braces to keep the scope local): tiny script footnote small normal large Large LARGE huge. Codes: `\tiny \scriptsize \footnotesize \small \normalsize \large \Large \LARGE \huge \Huge`.

3.3 Colour

```
{\color{red}text} → text \textcolor{blue}{text} → text
\colorbox{yellow}{hi} → hi \fcolorbox{black}{green!20}{hi} → hi
```

Define your own colour once, reuse it anywhere: `\definecolor{myblue}{RGB}{0,80,180}` then `\textcolor{myblue}{...}`. The `!` operator mixes: `blue!30` means 30 % blue, 70 % white.

4 Lists and Nested Lists

Three base environments: `itemize` (bullets), `enumerate` (numbers), `description` (bold terms). The `enumitem` package lets you set the label with `[label=...]` and reduce spacing with `[noitemsep]`.

```
\begin{enumerate}[label=(\alph*)]
  \item first
  \item second
    \begin{itemize}[label=$\star$]
      \item star bullet
      \item[$\oplus$] one-off marker
    \end{itemize}
\end{enumerate}
```

(a) first

(b) second

★ star bullet

⊕ one-off marker

	Produces	label=	Produces	label=
Common label patterns:	1. 2. 3.	<code>\arabic*.</code>	a) b) c)	<code>\alph*</code>
	(a) (b)	<code>(\alph*)</code>	I. II. III.	<code>\Roman*.</code>
	i. ii. iii.	<code>\roman*.</code>	A) B)	<code>\Alph*</code>
	†	<code>\$\dagger\$</code>	▷	<code>\$\triangleright\$</code>

The `description` list (definition style):

```
\begin{description}
  \item[Term A] its meaning.
  \item[Term B] its meaning.
\end{description}
```

Term A its meaning.

Term B its meaning.

Useful keys: `resume` (continue a previous list's count), `start=3` (begin at 3), `itemsep=2pt`, `leftmargin=1cm`.

5 Mathematics

5.1 The two modes

Inline math sits in the running line with `$...$`: the mass energy relation $E = mc^2$. *Display* math is centred on its own line with `\[...\]` (never the obsolete `$$...$$`):

$$\int_0^{\infty} e^{-x} dx = 1.$$

5.2 Numbered and multi-line equations

Environment	Purpose
<code>equation</code>	one numbered line
<code>align</code>	many lines, <code>&</code> = alignment column, each numbered
<code>align*</code>	same but no numbers
<code>aligned</code>	align inside ONE <code>equation</code> (single number)
<code>gather</code>	each line centred with its own number (no <code>&</code>)
<code>multline</code>	one long formula broken across lines
<code>cases</code>	piecewise definitions
<code>subequations</code>	wraps lines to number them (1a),(1b)

```
\begin{align}
  f(x) &= (x+1)^2 \\
  &= x^2 + 2x + 1
\end{align}
```

$$f(x) = (x + 1)^2 = x^2 + 2x + 1$$

Suppress one line's number with `\nonumber`; force a custom number with `\tag{$\ast$}`. The `&` always marks where lines align; `\\` ends a line.

5.3 Fractions, roots and big operators

Out	Code	Out	Code
$\frac{a}{b}$	<code>\dfrac{a}{b}</code>	$\sqrt{x}$	<code>\sqrt{x}</code>
$\frac{a}{b}$	<code>\tfrac{a}{b}</code>	$\sqrt[n]{x}$	<code>\sqrt[n]{x}</code>
$\binom{n}{k}$	<code>\binom{n}{k}</code>	x^{2n}	<code>x^{2n}</code>
$\sum_{i=1}^n$	<code>\sum_{i=1}^n</code>	$\int_a^b$	<code>\int_a^b</code>
$\prod_k$	<code>\prod_k</code>	$\lim_{x \rightarrow \infty}$	<code>\lim_{x \rightarrow \infty}</code>

5.4 Subscripts, superscripts and deep nesting

Always **group with braces**. e^{-st} breaks; $e^{\{-st\}}$ works. Nest freely: $x_{i_{j_k}}$; $e^{-t/(RC)}$ gives $e^{-t/(RC)}$; a symbol with both indices, $\Gamma_{\alpha\beta}^{\mu}$ gives $\Gamma_{\alpha\beta}^{\mu}$; a tensor with a gap, $A^{\mu}_{\ \nu}$ gives $A^{\mu}_{\ \nu}$. A stacked multi-line subscript uses `\substack`:

$$\sum_{\substack{i=1 \\ i \neq j}}^n a_{ij}.$$

5.5 Delimiters that grow

`\left...\right` auto-sizes to the content; they must be paired (use `\right.` for an invisible partner).

```
\left(\dfrac{a}{b}\right) \rightarrow \left(\frac{a}{b}\right) \left[\right] \rightarrow \left[\right] \left\{\right\} \rightarrow \left\{\right\}
\left|\right| \rightarrow || \leftangle\rightangle \rightarrow \langle \rangle
```

Manual sizes (small→big): `\big \Big \bigg \Bigg`. Floor/ceiling: `\lfloor x \rfloor` `\rceil` `\lceil x \rceil` `\rceil` `\lceil x \rceil`.

5.6 Matrices and cases

```
\begin{pmatrix} a & b \\ c & d \end{pmatrix}
\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}
f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}
```

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

Bracket styles by name: `pmatrix=( )`, `bmatrix=[ ]`, `Bmatrix={ }`, `vmatrix=| |`, `Vmatrix=|| ||`, `matrix=none`.

5.7 Spacing, text, braces and boxed answers

Spaces: thin `\,`, medium `\:`, thick `\;`, negative `\!`, and `\quad/\!\!\quad` for big gaps. Upright words in math: `\text{if } x > 0`. Annotate with braces: `\underbrace{x^2+2x+1}_{\text{square}}`

$= \underbrace{x^2 + 2x + 1}_{\text{square}}$ and $\overbrace{a+b}^n = a + b$. Box a final result: $\boxed{x = \frac{-b \pm \sqrt{\Delta}}{2a}}$.

6 Symbol Reference (glyph + code)

Every entry shows the printed symbol and the command that makes it. Most need `amsmath`/`amssymb` (already loaded here).

6.1 Lowercase Greek

α	<code>\alpha</code>	β	<code>\beta</code>	γ	<code>\gamma</code>	δ	<code>\delta</code>
ϵ	<code>\epsilon</code>	ε	<code>\varepsilon</code>	ζ	<code>\zeta</code>	η	<code>\eta</code>
θ	<code>\theta</code>	ϑ	<code>\vartheta</code>	ι	<code>\iota</code>	κ	<code>\kappa</code>
λ	<code>\lambda</code>	μ	<code>\mu</code>	ν	<code>\nu</code>	ξ	<code>\xi</code>
π	<code>\pi</code>	ϖ	<code>\varpi</code>	ρ	<code>\rho</code>	ϱ	<code>\varrho</code>
σ	<code>\sigma</code>	ς	<code>\varsigma</code>	τ	<code>\tau</code>	υ	<code>\upsilon</code>
ϕ	<code>\phi</code>	φ	<code>\varphi</code>	χ	<code>\chi</code>	ψ	<code>\psi</code>
ω	<code>\omega</code>	$\digamma$	<code>\digamma</code>	$\varkappa$	<code>\varkappa</code>		

6.2 Uppercase Greek

(Letters that look like Latin — A, B, E, Z, H, I, K, M, N, O, P, T, X — are just typed as cap-

Γ	<code>\Gamma</code>	Δ	<code>\Delta</code>	Θ	<code>\Theta</code>	Λ	<code>\Lambda</code>
Ξ	<code>\Xi</code>	Π	<code>\Pi</code>	Σ	<code>\Sigma</code>	Υ	<code>\Upsilon</code>
Φ	<code>\Phi</code>	Ψ	<code>\Psi</code>	Ω	<code>\Omega</code>	$\aleph$	<code>\aleph</code>

Slanted variants (with `amsmath`):

`\varGamma` `\varDelta` ... `\varGamma` `\varDelta` `\varTheta` `\varPhi` `\varOmega`.

6.3 Binary operators

$\pm$	<code>\pm</code>	$\mp$	<code>\mp</code>	$\times$	<code>\times</code>	$\div$	<code>\div</code>
$\cdot$	<code>\cdot</code>	$*$	<code>\ast</code>	$\star$	<code>\star</code>	$\circ$	<code>\circ</code>
$\bullet$	<code>\bullet</code>	$\oplus$	<code>\oplus</code>	$\ominus$	<code>\ominus</code>	$\otimes$	<code>\otimes</code>
$\oslash$	<code>\oslash</code>	$\odot$	<code>\odot</code>	$\cup$	<code>\cup</code>	$\cap$	<code>\cap</code>
$\uplus$	<code>\uplus</code>	$\sqcup$	<code>\sqcup</code>	$\sqcap$	<code>\sqcap</code>	$\wedge$	<code>\wedge</code>
$\vee$	<code>\vee</code>	$\setminus$	<code>\setminus</code>	$\wr$	<code>\wr</code>	$\diamond$	<code>\diamond</code>
$\bigtriangleup$	<code>\bigtriangleup</code>	$\bigtriangledown$	<code>\bigtriangledown</code>	$\triangleleft$	<code>\triangleleft</code>	$\triangleright$	<code>\triangleright</code>
$\dagger$	<code>\dagger</code>	$\ddagger$	<code>\ddagger</code>	$\amalg$	<code>\amalg</code>	$\boxplus$	<code>\boxplus</code>
$\boxtimes$	<code>\boxtimes</code>	$\boxdot$	<code>\boxdot</code>	$\ltimes$	<code>\ltimes</code>	$\rtimes$	<code>\rtimes</code>

6.4 Relations

$\leq$	<code>\leq</code>	$\geq$	<code>\geq</code>	$\neq$	<code>\neq</code>	$\equiv$	<code>\equiv</code>
$\approx$	<code>\approx</code>	$\cong$	<code>\cong</code>	$\simeq$	<code>\simeq</code>	$\sim$	<code>\sim</code>
$\propto$	<code>\propto</code>	$\ll$	<code>\ll</code>	$\gg$	<code>\gg</code>	$\asymp$	<code>\asymp</code>
$\doteq$	<code>\doteq</code>	$\prec$	<code>\prec</code>	$\succ$	<code>\succ</code>	$\preceq$	<code>\preceq</code>
$\succeq$	<code>\succeq</code>	$\subset$	<code>\subset</code>	$\supset$	<code>\supset</code>	$\subseteq$	<code>\subseteq</code>
$\supseteq$	<code>\supseteq</code>	$\sqsubset$	<code>\sqsubset</code>	$\sqsupseteq$	<code>\sqsupseteq</code>	$\in$	<code>\in</code>
$\ni$	<code>\ni</code>	$\notin$	<code>\notin</code>	$\vdash$	<code>\vdash</code>	$\dashv$	<code>\dashv</code>
$\models$	<code>\models</code>	$\perp$	<code>\perp</code>	$\mid$	<code>\mid</code>	$\parallel$	<code>\parallel</code>
$\bowtie$	<code>\bowtie</code>	$\smile$	<code>\smile</code>	$\frown$	<code>\frown</code>	$\therefore$	<code>\therefore</code>
$\because$	<code>\because</code>	$\trianglelefteq$	<code>\trianglelefteq</code>	$\lesssim$	<code>\lesssim</code>	$\gtrsim$	<code>\gtrsim</code>

6.5 Negated relations

$\neq$	<code>\neq</code>	$\notin$	<code>\notin</code>	$\nleq$	<code>\nleq</code>	$\ngeq$	<code>\ngeq</code>
$\nless$	<code>\nless</code>	$\ngtr$	<code>\ngtr</code>	$\nsim$	<code>\nsim</code>	$\ncong$	<code>\ncong</code>
$\nsubseteq$	<code>\nsubseteq</code>	$\nsupseteq$	<code>\nsupseteq</code>	$\subsetneq$	<code>\subsetneq</code>	$\supsetneq$	<code>\supsetneq</code>
$\nprec$	<code>\nprec</code>	$\nsucc$	<code>\nsucc</code>	$\nmid$	<code>\nmid</code>	$\nparallel$	<code>\nparallel</code>
$\nvdash$	<code>\nvdash</code>	$\nvDash$	<code>\nvDash</code>	$\nleftarrow$	<code>\nleftarrow</code>	$\nrightarrow$	<code>\nrightarrow</code>

6.6 Arrows

$\leftarrow$	<code>\leftarrow</code>	$\rightarrow$	<code>\rightarrow</code>
$\leftrightarrow$	<code>\leftrightarrow</code>	$\Leftarrow$	<code>\Leftarrow</code>
$\Rightarrow$	<code>\Rightarrow</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\longleftarrow$	<code>\longleftarrow</code>	$\longrightarrow$	<code>\longrightarrow</code>
$\longleftrightarrow$	<code>\longleftrightarrow</code>	$\Longrightarrow$	<code>\Longrightarrow</code>
$\Longleftrightarrow$	<code>\Longleftrightarrow</code>	$\mapsto$	<code>\mapsto</code>
$\longmapsto$	<code>\longmapsto</code>	$\hookleftarrow$	<code>\hookleftarrow</code>
$\hookrightarrow$	<code>\hookrightarrow</code>	$\uparrow$	<code>\uparrow</code>
$\downarrow$	<code>\downarrow</code>	$\updownarrow$	<code>\updownarrow</code>
$\Uparrow$	<code>\Uparrow</code>	$\Downarrow$	<code>\Downarrow</code>
$\nearrow$	<code>\nearrow</code>	$\searrow$	<code>\searrow</code>
$\swarrow$	<code>\swarrow</code>	$\nwarrow$	<code>\nwarrow</code>
$\rightleftharpoons$	<code>\rightleftharpoons</code>	$\leftharpoonup$	<code>\leftharpoonup</code>
$\rightharpoonup$	<code>\rightharpoonup</code>	$\twoheadrightarrow$	<code>\twoheadrightarrow</code>
$\rightarrowtail$	<code>\rightarrowtail</code>	$\leadsto$	<code>\leadsto</code>
$\rightrightarrows$	<code>\rightrightarrows</code>	$\rightleftarrows$	<code>\rightleftarrows</code>

$\xrightarrow{f}, \xleftarrow{g}$.

Labelled arrows (need `amsmath`): `\xrightarrow{f}`

6.7 Big (variable-size) operators

$\sum$	<code>\sum</code>	$\prod$	<code>\prod</code>	$\coprod$	<code>\coprod</code>	$\int$	<code>\int</code>
$\oint$	<code>\oint</code>	$\iint$	<code>\iint</code>	$\iiint$	<code>\iiint</code>	$\bigcup$	<code>\bigcup</code>
$\bigcap$	<code>\bigcap</code>	$\bigsqcup$	<code>\bigsqcup</code>	$\bigvee$	<code>\bigvee</code>	$\bigwedge$	<code>\bigwedge</code>
$\bigoplus$	<code>\bigoplus</code>	$\bigotimes$	<code>\bigotimes</code>	$\bigodot$	<code>\bigodot</code>	$\biguplus$	<code>\biguplus</code>

6.8 Delimiters

$\langle$	<code>\langle</code>	$\rangle$	<code>\rangle</code>	$\lfloor$	<code>\lfloor</code>	$\rfloor$	<code>\rfloor</code>
$\lceil$	<code>\lceil</code>	$\rceil$	<code>\rceil</code>	$\ $	<code>\ </code>	$\ $	<code>\ </code>
$\ $	<code>\ </code>	$\backslash$	<code>\backslash</code>	$\ulcorner$	<code>\ulcorner</code>	$\urcorner$	<code>\urcorner</code>

6.9 Dots and accents

$\dots$	<code>\dots</code>	$\cdots$	<code>\cdots</code>	$\vdots$	<code>\vdots</code>	$\ddots$	<code>\ddots</code>
$\hat{a}$	<code>\hat{a}</code>	$\bar{a}$	<code>\bar{a}</code>	$\tilde{a}$	<code>\tilde{a}</code>	$\vec{a}$	<code>\vec{a}</code>
$\dot{a}$	<code>\dot{a}</code>	$\ddot{a}$	<code>\ddot{a}</code>	$\acute{a}$	<code>\acute{a}</code>	$\grave{a}$	<code>\grave{a}</code>
$\breve{a}$	<code>\breve{a}</code>	$\check{a}$	<code>\check{a}</code>	$\ddot{a}$	<code>\ddot{a}</code>	$\widehat{abc}$	<code>\widehat{abc}</code>
$\widetilde{abc}$	<code>\widetilde{abc}</code>	$\overline{abc}$	<code>\overline{abc}</code>	$\underline{abc}$	<code>\underline{abc}</code>	$\overrightarrow{abc}$	<code>\overrightarrow{abc}</code>

6.10 Miscellaneous symbols

∞	<code>\infty</code>	∂	<code>\partial</code>	∇	<code>\nabla</code>	$\emptyset$	<code>\emptyset</code>
$\varnothing$	<code>\varnothing</code>	$\forall$	<code>\forall</code>	$\exists$	<code>\exists</code>	$\nexists$	<code>\nexists</code>
$\neg$	<code>\neg</code>	$\top$	<code>\top</code>	$\bot$	<code>\bot</code>	$\angle$	<code>\angle</code>
$\measuredangle$	<code>\measuredangle</code>	$\triangle$	<code>\triangle</code>	$\square$	<code>\square</code>	$\blacksquare$	<code>\blacksquare</code>
$\blacktriangle$	<code>\blacktriangle</code>	$\lozenge$	<code>\lozenge</code>	$\blacklozenge$	<code>\blacklozenge</code>	$\eth$	<code>\eth</code>
$\clubsuit$	<code>\clubsuit</code>	$\diamondsuit$	<code>\diamondsuit</code>	$\heartsuit$	<code>\heartsuit</code>	$\spadesuit$	<code>\spadesuit</code>
$\flat$	<code>\flat</code>	$\natural$	<code>\natural</code>	$\sharp$	<code>\sharp</code>	$\Re$	<code>\Re</code>
$\Im$	<code>\Im</code>	$\wp$	<code>\wp</code>	ℓ	<code>\ell</code>	$\hbar$	<code>\hbar</code>
$\imath$	<code>\imath</code>	$\jmath$	<code>\jmath</code>	$\aleph$	<code>\aleph</code>	$\complement$	<code>\complement</code>
$\prime$	<code>\prime</code>	$\surd$	<code>\surd</code>	$\checkmark$	<code>\checkmark</code>	$\maltese$	<code>\maltese</code>
$\textcircled{R}$	<code>\textcircled{R}</code>	$\textcircled{S}$	<code>\textcircled{S}</code>	$\bigstar$	<code>\bigstar</code>	$\mho$	<code>\mho</code>

6.11 Special alphabets

Output	Code
$\mathbb{R} \mathbb{N} \mathbb{Z} \mathbb{Q} \mathbb{C}$	<code>\mathbb{R} \mathbb{N} \mathbb{Z} \mathbb{Q} \mathbb{C}</code> ... (blackboard, amssymb)
$\mathcal{A} \mathcal{B} \mathcal{L} \mathcal{F}$	<code>\mathcal{A} \mathcal{B} \mathcal{L} \mathcal{F}</code> ... (calligraphic)
$\mathscr{A} \mathscr{B} \mathscr{L} \mathscr{F}$	<code>\mathscr{A} \mathscr{B} \mathscr{L} \mathscr{F}</code> ... (script, mathrsfs)
$\mathfrak{A} \mathfrak{g} \mathfrak{p} \mathfrak{m}$	<code>\mathfrak{A} \mathfrak{g} \mathfrak{p} \mathfrak{m}</code> ... (fraktur, amssymb)
$\mathbf{A} \mathbf{A} \mathbf{A} \mathbf{A} \mathbf{A}$	<code>\mathbf{A} \mathbf{A} \mathbf{A} \mathbf{A} \mathbf{A}</code> <code>\mathrm</code> <code>\mathsf</code> <code>\mathtt</code> <code>\bm</code>

6.12 Named functions / operators

Type these as commands so they print upright with correct spacing: `\sin \cos \tan \cot \sec \csc \arcsin \cosh \tanh \coth \exp \log \ln \lg \lim \limsup \liminf \sup \inf \max \min \arg \deg \det`. For a *custom* operator with limits use `\DeclareMathOperator*{\argmax}{arg\,max}` in the preamble.

6.13 Text-mode symbols (outside math)

$\dagger$ <code>\dag</code>	$\ddagger$ <code>\ddag</code>	$\S$ <code>\S</code>	$\P$ <code>\P</code>	
$\copyright$ <code>\copyright</code>	$\textregistered$ <code>\textregistered</code>	TM <code>\texttrademark</code>	$\pounds$ <code>\pounds</code>	
$\bullet$ <code>\textbullet</code>	$^\circ$ <code>\textdegree</code>	$\backslash$ <code>\textbackslash</code>	$\ldots$ <code>\ldots</code>	Check/cross via
$\&$ <code>\&</code>	$\%$ <code>\%</code>	$\#$ <code>\#</code>	$_$ <code>_</code>	

pifont: `\ding{51}` ✓ `\ding{55}` ✗.

7 Tables

7.1 The basics

A `tabular` needs a *column spec*: `l/c/r` (left/centre/right), `|` for a vertical rule, `p{3cm}` for a fixed wrapping width. Inside: `&` separates columns, `\\` ends a row, `\hline` draws a full horizontal rule, `\cline{2-3}` a partial one.

```
\begin{tabular}{|l|c|r|}
\hline
Left & Mid & Right\\
a & b & c\\
\hline
\end{tabular}
```

Left	Mid	Right
a	b	c

7.2 multirow, multicolumn, colour and math in cells

`\multirow{N}{*}{txt}` spans N rows; `\multicolumn{N}{c|txt|}` spans N columns (and *re-declares* the borders). Colour with `\rowcolor` (must be the **first** token of the row), `\cellcolor`, `\columncolor`. Put maths in a cell with $\$...\$$.

```

\renewcommand{\arraystretch}{1.4}
\begin{tabular}{|c|c|c|}
\hline
\rowcolor{gray!30}
\multicolumn{2}{|c|}{\textbf{Header}} & C\\
\hline
\multirow{2}{*}{Merged} &  $x^2$  &  $s > 0$  \\
&  $\frac{1}{s}$  &  $s > 0$  \\
\hline
\end{tabular}

```

Header		C
Merged	x^2	$s > 0$
	$\frac{1}{s}$	$s > 0$

Taller rows: `\renewcommand{\arraystretch}{1.5}`. Thicker rules: `\setlength{\arrayrulewidth}{1pt}` or use `booktabs` (`\toprule` `\midrule` `\bottomrule`, no vertical rules — the modern look).
Line breaks inside a cell: `\makecell{line 1\\ line 2}` (`makecell`).

7.3 Auto-width tables (tabularx)

When a column must stretch to fill a set total width, use `tabularx` and the `X` column type:

```
\begin{tabularx}{0.8\textwidth}{|l|X|}
```

Key	This X-column wraps automatically to fill the declared width so the table always spans exactly 80% of the text width.
-----	---

8 Figures and Graphics

8.1 One figure

```

\begin{figure}[htbp] % h=here t=top b=bottom p=floats-page
\centering
\includegraphics[width=0.5\textwidth]{example-image-a}
\caption{A single figure.} % \caption BEFORE \label
\label{fig:single}
\end{figure}

```

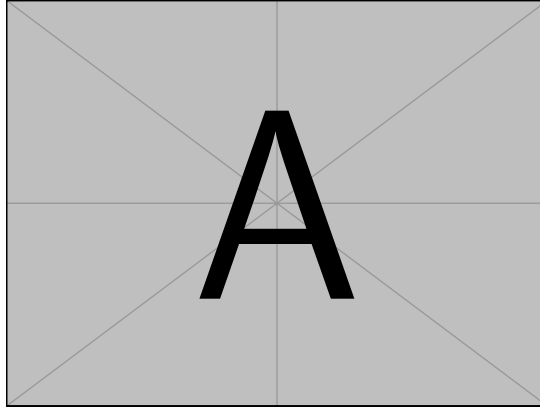


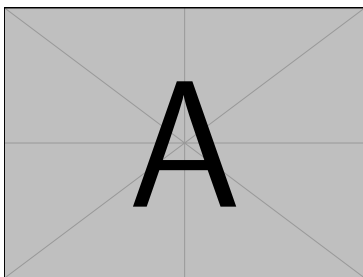
Figure 1: A single figure (rendered here with [H] from float).

Scaling keys for `\includegraphics`: `width=`, `height=`, `scale=0.5`, `angle=90` (rotate), `keepaspectratio`. Force an exact position with [H] (needs float); use [htbp!] to strongly suggest.

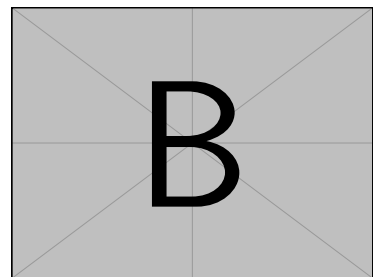
8.2 Side-by-side subfigures

`subcaption` gives each panel its own (a),(b) sub-label.

```
\begin{figure}[htbp]\centering
  \begin{subfigure}[b]{0.3\textwidth}
    \includegraphics[width=\linewidth]{example-image-a}
    \caption{First}\label{fig:a}
  \end{subfigure}\hfill
  \begin{subfigure}[b]{0.3\textwidth}
    \includegraphics[width=\linewidth]{example-image-b}
    \caption{Second}\label{fig:b}
  \end{subfigure}
  \caption{Two panels.}\label{fig:pair}
\end{figure}
```



(a) First



(b) Second

Figure 2: Two panels (rendered).

`\ref{fig:a}` gives the full number “??”; `\subref{fig:a}` gives just the letter. For panels *without* sub-captions, use `\begin{minipage}{0.48\textwidth}...\end{minipage}` instead.

8.3 Wrapped and rotated figures

```
\begin{wrapfigure}{r}{0.35\textwidth} % r=right side, l=left side
  \centering
```

```

\includegraphics[width=\linewidth]{example-image}
\caption{Text flows around me.}\label{fig:wrap}
\end{wrapfigure}
% ---- rotation ----
\includegraphics[angle=90]{example-image} % rotate the image
\rotatebox{45}{slanted text} % rotate anything (graphicx)
\begin{sidewaysfigure}...\end{sidewaysfigure} % 90-deg whole float (rotating)

```

Rotated text sample:  .

9 Cross-References and Hyperlinks

	Command	Prints
	<code>\ref{k}</code>	the bare number
	<code>\eqref{k}</code>	(number)
Label anything right after its <code>\caption</code> (or in an equation), then point at it:	<code>\pageref{k}</code>	the page number
	<code>\autoref{k}</code>	“Figure 3”
	<code>\cite{k}</code>	[1]

External links (`\hyperref`): clickable text `\href{https://example.com}{visit site}` → [visit site](#); bare URL `\url{https://example.com}`. Always put a `~` (non-breaking space) before a reference so the number never wraps alone: Figure~`\ref{fig:pair}`. Configure link colours once with `\hypersetup{colorlinks=true,...}`.

10 Code Listings

The `listings` package typesets source code verbatim with optional syntax colouring and line numbers.

Listing 1: Inline Python

```

1 def rc_response(R, C, V, times):
2     """One-line docstring."""
3     tau = R * C
4     return [V * (1 - 2.718**(-t / tau)) for t in times]

```

Inline snippet: `\lstinline!print(x)!`. From an external file: `\lstinputlisting[style=py]{script.py}`. Define a reusable style once with `\lstdefinestyle{py}{language=Python, numbers=left, frame=single, keywordstyle=\color{blue}, ...}`. For publication-grade output the `minted` package is prettier but needs `-shell-escape` + Python.

11 Theorems, Proofs and Coloured Boxes

Declare environments in the preamble with `amsthm`: `\newtheorem{theorem}{Theorem}[section]` numbers them “1.1, 1.2”.

Definition 11.1. A *prime* is an integer > 1 divisible only by 1 and itself.

Theorem 11.1 (Pythagoras). In a right triangle, $a^2 + b^2 = c^2$.

Proof. Standard area argument. □

The `proof` environment adds the italic “Proof.” lead-in and the □. A *coloured, titled* theorem box via `tcolorbox`:

Key Result 11.1: Weak-field orbit

$$\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}.$$

Reference them like anything else: Theorem ??.

12 Algorithms / Pseudocode

Algorithm 1: Find the maximum

Input: array A of length n

Output: maximum element

```
1  $m \leftarrow A[1]$ ;
2 for  $i \leftarrow 2$  to  $n$  do
3   | if  $A[i] > m$  then
4   |   |  $m \leftarrow A[i]$ ;
5   | end
6 end
7 return  $m$ ;
```

Using `algorithm2e` (loaded with `[ruled,vlined,linesnumbered]`):

Key commands: `\KwIn \KwOut \For \While \If \ElseIf \Else \Return`, end each statement line with `\;`. The alternative package pair is `algorithm + algpseudocode` (`\State \For \EndFor`).

13 Footnotes and Bibliography

A footnote¹ is placed with `\footnote{...}` immediately after the word. The fastest bibliography for a timed exam is the manual `thebibliography` environment (no BibTeX run needed):

```
\begin{thebibliography}{9}    % {9} = widest label is one digit
  \bibitem{kreyszig} E. Kreyszig, \textit{Advanced Engineering
    Mathematics}, 10th ed. Wiley, 2011.
  \bibitem{ogata} K. Ogata, \textit{Modern Control Engineering},
    5th ed. Prentice Hall, 2010.
\end{thebibliography}
```

Cite in the text with `\cite{kreyszig}`, which prints [1] and links to the entry.

14 Predicted Tricky Variations

1. **Multi-column aligned maths** — when two equation pairs sit side by side, align spaces them oddly; use `alignat`, where you control the gaps: `\begin{alignat}{2} a\&=b \&\quad c\&=d \end{alignat}`
2. **Numbered sub-equations** (1a),(1b): wrap an `align` in `\begin{subequations}...\end{subequations}`.

¹This text drops to the bottom of the page and auto-numbers.

3. **Numbered derivation, one number** — keep `&=` aligned and end each intermediate line with `\nonumber` so only the last is numbered.
4. **Clean scientific notation / units** — `siunitx`: `\num{1.496e11}` $\rightarrow$ 1.496×10^{11} and `\SI{9.8}{m/s^2}`.
5. **Coloured full-row header** — `\rowcolor` must be the FIRST token of the row, even before `\multicolumn`.
6. **Landscape float** — `sidewaystable`/`sidewaysfigure` (rotating) or the `{landscape}` environment (`lscape`).
7. **Custom numbered coloured theorem** — `tcolorbox` with `\newtcbtheorem` (as the “Key Result” box above).
8. **Tensor / pre-indices** — `\{}_a^b` X for pre-scripts, and a phantom space `\` (or `\vphantom`) to stop indices stacking.
9. **Continued fractions** — `\cfrac{1}{1+\cfrac{1}{1+x}}`.
10. **Highlight/box a final answer** — `\boxed{...}` or `\colorbox{yellow}{$...$}`.

15 Common Errors and Instant Fixes

Message / symptom	Fix
Missing \$ inserted	A maths symbol used in text — wrap it in <code>\$...\$</code> .
Undefined control sequence	Typo, or the package that defines it isn’t loaded.
! Misplaced alignment tab & <code>\ref</code> shows “??”	A bare <code>&</code> outside a table/align — escape it <code>\&</code> . Compile again (labels resolve on the 2nd pass).
Float “lost” / on wrong page	Use <code>[htbp]</code> , add <code>!</code> , or <code>[H]</code> (<code>float</code>).
Table too wide	Shrink columns, use <code>tabularx</code> , or <code>\small</code> / <code>\resizebox</code> .
Fractions collide in a table	<code>\renewcommand{\arraystretch}{1.5}</code> .
Option clash for package <code>xcolor</code>	Load <code>xcolor</code> (with options) <i>before</i> <code>tcolorbox/tikz</code> .
Accented letter breaks	Load <code>\usepackage[T1]{fontenc}</code> and <code>[utf8]{inputenc}</code> .

16 Exam Speed Strategy

1. Paste a preamble first; add packages only as each feature appears.
2. Build in reading order: preamble $\rightarrow$ title $\rightarrow$ sections/lists $\rightarrow$ equations $\rightarrow$ tables $\rightarrow$ figures $\rightarrow$ code $\rightarrow$ references.
3. **Compile every few environments** so any error is easy to locate.
4. **Copy–paste–tweak** repetitive blocks (table rows, subfigures, `\bibitems`) — never retype them.

5. Keep this compiled PDF open on a second window for instant symbol lookup.
6. Leave line breaks, exact float positions and page breaks for last — they are usually *not* graded, but successful compilation always is.

End of reference — compile twice, print, and keep beside you.